

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 1 of 10

(1) (1) Research Program/Project/Study Title: Cured Semi-Dried (Hubag-hubag) Boneless Bangus: Potential for Commercialization	
Proponent Name:	JANET LEDESMA
Academic Rank:	Instructor 1
Co-Authors (with academic rank):	LOUELLA P. DADIVAS Instructor 1
Campus/Satellite College/Address:	CAPSU DAYAO SATELLITE COLLEGE
Telephone/Cell Phone/Fax Number:	
E-mail Address:	jvledesma@capsu.edu.com

Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	ROXAS CITY
		Province:	CAPIZ
		Region:	VI
(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity		(8) Discipline/Field of Specialization Foods	

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 2 of 10

(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The researcher came up with the idea of “Cured Semi-Dried (Hubag-hubag) Boneless Bangus: Potential for Commercialization. Bangus, also known as milkfish, (scientific name: *Chanos Chanos*) is most closely related to carp and catfishes. It occurs in the Indian Ocean and across the Pacific Ocean, tending to schools around coasts and islands with reefs. In the Philippines, bangus can be raised anywhere. This study was conducted to determine the acceptability of “Dried Flavored Boneless Bangus (Potential for Commercialization)” Specifically, it aims to: (1) Determine the sensory qualities of dried flavored boneless Bangus. (2) Determine the consumers’ preferred flavor of dried boneless Bangus (3) Determine the significant difference in the acceptability of the four treatments.

The experimental comparative study design was used in this study using the analysis- of variance (ANOVA) in determining the significant differences.

In this study, it was found that dried boneless Bangus with Kamias flavor) had the highest sensory quality among the four treatments in terms of appearance, texture, and taste, but Treatment A (with Kamias flavor) and Treatment C (with Batwan flavor) had both the highest in terms of aroma. Treatment A had also the highest consumer acceptability, although there were no significant differences that exist among the treatments considering all the variables such as appearance, texture, taste, and aroma.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Indigenous souring agents in the Philippines are very helpful not as souring agents but have nutrition values that help to improve the body's immune. Labog (roselle) has several properties as a medicinal plant since it is used as a diuretic, antibilious and antiscorbutic, digestive, laxative, choleric, and hypotensive. Furthermore, its seed contains fixed oil (Khare, 2007). Libas is a deciduous tree cultivated in many Asian countries. Raw fruits of libas are eaten fresh or pickled. Phytochemical studies have revealed the presence of various constituents such as phenolic acids, flavonoids, amino acids, and minerals which are of great importance for human health. Various extracts of fruit and leaves of kami (Averrhoa bilimbi) have anti-diabetic, anti-microbial, anti-inflammatory, cytotoxic activities, anti-oxidant activity, anti-fertility, and anti-bacterial activities. (P. Vazhacharickal 2017). Batuan an indigenous, lesser-known member of the Guttiferae family with promising export potential is understood and underutilized.

Drying is the removal of water from fish. Normally the term 'drying' implies the removal of water by evaporation but water can be removed by other methods: for example, the action of salt and the application of pressure will remove water from fish. During air drying, water is removed from the surface of the fish, and water moves from the deeper layers to the surface.

In this study, the researcher aims to produce nutritious food which is organic indigenous flavors such as Kamias, Libas, Batwan, and Labog for boneless bangus in a convenient way because of the absence of numerous bones in fish, to know if this will be popular with everyone especially kids to prevent eating bones from fish also the researcher wants to know if this will be known for its distinct flavor and aimed to become a potential for commercialization. Since Bangus is an available product of CapSU Dayao and Boneless Bangus is the usual product, they sold in the market and not the flavored ones. The researcher also aims to develop a value-added product for boneless Bangus.

This study was conducted to determine the acceptability of “Dried Flavored Boneless Bangus (Potential for Commercialization)” Specifically, it aims to

1. Determine the sensory qualities of dried flavored boneless Bangus.
2. Determine the consumers’ preferred flavor dried boneless bangus.
3. Determine the significant difference in the acceptability of the four treatments.

(11) Materials and Methods or Methodology

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 3 of 10

(This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science research and methodology in Social Sciences. Presentation of materials may not necessarily be in a separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Materials and Methods

Materials.

The following are the materials used in the preparation of cured semi-dried boneless bangus: knife, chopping board, forceps preparation tray, dehydrator, measuring spoon, measuring cups, weighing scale, mixing bowl, tong, wooden spoon.

Raw Materials.

The following raw materials used in the preparation of cured semi-dried boneless bangus: bangus, libas, batwan, labog kamias, salt, sugar, water, garlic, black pepper.

Preparation of Boneless Bangus with flavor

The Bangus will be washed, and split fish into butterfly fillets. Remove the internal organs. Wash thoroughly. Remove the backbone and dorsal fin. Make a line with the use of a knife/forceps, start removing the rib bones, and make a slit along with the vent of dorsal and ventral muscles with the edge of a knife. Remove the spines one after the other by inserting the pair of forceps between the segments of the dorsal and ventral muscles. Brine the boneless bangus with water, salt, and sugar solution for 30 minutes. After 30 minutes of brining, remove the boneless Bangus from brining solution. Cure the fish with the flavor you choose (Kamias, Libas, Batwan, Labog) garlic, and black pepper for 2 hours then dehydrate the cured boneless bangus in the dehydrator for 5 hrs at 35 °C. Set aside

Sensory Evaluation of Dried Flavored Boneless Bangus

The sensory evaluation consists of judging the quality of food by a panel of 10 subject experts of CapSU, and 100 consumers of Roxas City. The evaluation deals with analyzing and interpreting the qualities of the cured semi-dried boneless Bangus as they are perceived by the sense of sight, taste, and touch. The 9-point Hedonic scale score-card method was used to determine the sensory characteristics of cured semi-dried boneless Bangus. Sensory attributes of the cured semi-dried boneless Bangus namely, appearance, texture, taste, aroma, and overall acceptability will be allotted a maximum of 9 each. The scoring scale has a rating of 1 (dislike extremely), 2 (dislike very much), 3 (dislike moderately), 4 (dislike slightly), 5 (neither like or dislike), 6 (liked slightly), 7 (liked moderately), 8 (liked very much), 9 (liked extremely).

Experimental Design and Treatment

Experimental Design

The study used an experimental method of research. An experimental method is a method or procedure involving the control or manipulation of conditions for the purpose of studying the relative effects of various treatments applied to the mixture of different samples (Calderon, 2002)

Experimental Treatment

The study used four (4) treatments with three replications: Treatment A (250g Kamias.), Treatment B (250g, Libas), Treatment C (250g, Batwan), and Treatment D (250g, Labog). The same ingredients and proportion was used in the study.

For the sensory evaluation:

Cured Semi-Dried (Hubag-hubag) was prepared in accordance with the protocol for sensory evaluation.

Scorecards which contain a 9-point scale for the sensory characteristics and a nine-point Hedonic scale for acceptability were used to evaluate the developed product. 10 semi-trained evaluators evaluated the product.

Only sensory characteristics such as appearance, texture, taste, and aroma were evaluated.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 4 of 10

Data Collection

The data was gathered using a score containing the 9-point hedonic scale. It contains the rating of the panelist on appearance, texture, taste, aroma, and general acceptability. The data was gathered after every replication.

Scoring of Variables

The criteria for evaluation as to the sensory characteristics of the product was a 9-point hedonic scale in terms of color, odor, texture, and taste. The rating for general acceptability in a 9-point hedonic scale was: 9 for like extremely, 8 for like very much, 7 for like moderately, 6 for like slightly, 5 for neither like nor a dislike, 4 for dislike moderately, 3 for dislike slightly, 2 for dislike very much and 1 for dislike extremely (Calmorin 2006).

The score on each item was interpreted based on the result of the computation as to aroma, texture, taste, appearance, and general acceptability using the following scoring interval and verbal interpretation.

For Aroma

Scoring	Verbal interpretation/ Response Categories	Scoring Interval
9	Extremely inviting	8.12 – 9.0
8	Very Inviting	7.23-8.11
7	Moderately Inviting	6.34-7.22
6	Slightly Inviting	5.45-6.33
5	Neither Inviting Nor not inviting	4.56-5.44
4	Slightly Not inviting	3.67-4.55
3	Moderately not inviting	2.78-3.66
2	Very not inviting	1.89-2.77
1	Extremely not inviting	1.00-1.88

For Texture

Scoring	Verbal Interpretation/ Response Categories	Scoring Interval
9	Extremely Firm	8.12 – 9.0
8	Very Firm	7.23-8.11
7	Moderately Firm	6.34-7.22
6	Slightly Firm	5.45-6.33
5	Neither Firm nor Soft	4.56-5.44
4	Slightly Soft	3.67-4.55
3	Moderately Soft	2.78-3.66
2	Very Soft	1.89-2.77
1	Extremely Soft	1.00-1.88

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 5 of 10

For Taste

Scoring

9
8
7
6
5
4
3
2
1

Verbal interpretation/ Response Categories

Extremely Delicious
Very delicious
Moderately delicious
Slightly delicious
Neither delicious Nor not Delicious
Slightly Not Delicious
Moderately not delicious
Very not delicious
Extremely not delicious

Scoring Interval

8.12 – 9.0
7.23-8.11
6.34-7.22
5.45-6.33
4.56-5.44
3.67-4.55
2.78-3.66
1.89-2.77
1.00-1.88

For Appearance

Scoring

9
8
7
6
5
4
3
2
1

Verbal Interpretation/ Response Categories

Extremely Tan
Very Tan
Moderately Tan
Slightly Tan
Pale
Slightly Pale
Moderately Pale
Very Pale
Extremely Pale

Scoring Interval

8.12 – 9.0
7.23-8.11
6.34-7.22
5.45-6.33
4.56-5.44
3.67-4.55
2.78-3.66
1.89-2.77
1.00-1.88

For General Acceptability

Scoring

9
8
7
6
5
4
3
2
1

Score Interval

8.12 – 9.0
7.23 – 8.11
6.34 – 7.22
5.45 – 6.33
4.56 – 5.44
3.67 – 4.55
2.78 – 3.66
1.89 – 2.77
1.00 – 1.88

Verbal interpretation/Response Categories

Like Extremely
Like Very Much
Like Moderately
Like Slightly
Neither Like nor Dislike
Dislike Slightly
Dislike Moderately
Dislike Very Much
Dislike Extremely

Statistical Tools and Analysis

The data were analyzed using weighted mean and Analysis of Variance (ANOVA).

The weighted mean was used to determine the acceptability of the sensory characteristics such as aroma, texture, taste, appearance, and general acceptability of the product.

The analysis of variance was used to determine the significant difference among treatments in terms of aroma, texture, taste, appearance.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 6 of 10

For Moisture Content:

The sample was sent to DOST or any available accredited agency for analysis. The package was dependent on the requirement set by an accredited agency.

(12) Results and Discussions (This part contains the presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. The sequence shall be based on the objectives of the study. Tables are formatted without vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Results and Discussions

Sensory characteristics of Different treatments In terms in aroma, texture, taste, appearance, and general acceptability

Contained in **Table 1** are the sensory qualities of the four treatments of Cured Semi-dried Bangus with Kamias, Libas, Batwan, and Labog flavors. It was found that in terms of appearance, cured Semi-dried Bangus with Kamias flavor had the highest mean of 7.98 described as Very Tan, while Treatment D (with Labog Flavor) had the lowest with 7.37 described as Moderately Tan.

In terms of Texture, Treatment A had also the highest mean at 7.9 described as very firm, while Treatment D had the lowest at 7.43 described as Moderately Firm.

In terms of taste, the same Treatment A had the highest mean score of 7.53 described as Very Delicious, while the lowest was Treatment B (with Libas flavor) with 7.23 described as Moderately Delicious. Treatment C (with Labog flavor) and D (with Batwan Flavor) were also described as Moderately delicious even though they had higher mean scores.

In terms of aroma, Treatment A (with Kamias flavor) and C (with Batwan flavor) had the highest mean identical at 7.53 described as very inviting, while Treatment D (with Labog favor) had the lowest at 7.23 described as Moderately Inviting.

Table 1. Sensory qualities of Cured Semi-dried Bangus treatments with Kamias, Libas, Batwan, and Labog flavors.

Treatments	A	Description	B	Description	C	Description	D	Description
Texture	7.9	VF	7.6	VF	7.63	VF	7.43	MF
Taste	7.53	VD	7.23	MD	7.47	MD	7.3	MD
Aroma	7.53	VI	7.3	MI	7.53	VI	7.23	MI
General Acceptability	7.74	LVM	7.45	LM	7.59	LVM	7.33	LM

Legend:

VT – Very Tan
MT – Moderately Tan
VF – Brown
MF –Dark Brown
LVM – Like Very Much

VD – Very Delicious
MD – Moderately Delicious
VI – Very Inviting
MI – Moderately Inviting
LM - Like Moderately

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 7 of 10

Data indicate that the general acceptability of treatment A as perceived by the evaluators was 7.77 on the Hedonic Scale after three replications considering appearance, texture, taste, and aroma, interpreted as Like Very Much. The sensory qualities of Treatment A had the highest mean in terms of appearance and texture with 7.97, described as Very Tan. It had an identical lower mean of 7.53 in terms of taste and aroma, but nonetheless, it was also described as Very Tan.

Table 2. Sensory Qualities of Treatment A-Cured Semi-dried Bangus in Kamias Flavor Considering Appearance, Texture, Taste, Aroma, and Its General Acceptability

Treatment A	Replicate 1 2 3			Total	Mean	Verbal Interpretation
Appearance	8.1	8.1	7.7	23.9	7.97	Very Tan
Texture	7.8	8.1	7.8	23.7	7.9	Very Tan
Taste	7.5	7.7	7.4	14.9	7.45	Very Tan
Aroma	7.7	7.4	7.5	22.6	7.53	Very Tan
General Acceptability	7.8	7.7	7.8	23.3	7.77	Like Very Much

After three replications, Treatment B received a 7.6 general acceptability mean score from the evaluators, which was interpreted as Like Very Much. The sensory quality of the treatment in terms of appearance had a mean score of 7.67, described as Very Tan, while in terms of aroma, it had a mean score of 7.3 described as Moderately Inviting, and in terms of taste, it had 7.23 described as Moderately Delicious.

Table 3. Sensory Qualities of Treatment B-Cured Semi-dried Bangus in Libas Flavor Considering Appearance, Texture, Taste, Aroma, and Its General Acceptability

Treatment B	Replicate 1 2 3			Total	Mean	Verbal Interpretation
Appearance	7.9	7.6	7.5	23	7.67	Very Tan
Texture	7.8	7.3	7.7	22.8	7.6	Very Tan
Taste	7.3	7.2	7.2	14.5	7.25	Very Tan
Aroma	7.5	7.1	7.3	21.9	7.3	Moderately Inviting
General Acceptability	7.9	7.3	7.6	22.8	7.6	Like Very Much

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 8 of 10

After three replications, Treatment B received a 7.6 general acceptability mean score from the evaluators, which was interpreted as Like Very Much. The sensory quality of the treatment in terms of appearance had a mean score of 7.67, described as Very Tan, while in terms of aroma, it had a mean score of 7.3 described as Moderately Inviting, and in terms of taste, it had 7.23 described as Moderately Delicious.

Table 4. Sensory Qualities of Treatment C-Cured Semi-dried Bangus in Labog Flavor Considering Appearance, Texture, Taste, Aroma, and Its General Acceptability

Treatment C	Replicate			Total	Mean	Verbal Interpretation
	1	2	3			
Appearance	7.3	7.8	8.1	23.2	7.67	Very Tan
Texture	7.3	7.8	7.8	22.9	7.6	Like Very Much
Taste	7.4	7.5	7.5	14.9	7.23	Moderately Delicious
Aroma	7.4	7.3	7.9	22.6	7.5	Moderately Inviting
General Acceptability	7.5	7.7	7.7	22.9	7.6	Like Very Much

The data disclosed that Treatment D was Liked Moderately as shown by the mean score of 7.33. Pertaining to its sensory qualities, the treatment was described as Moderately Tan having a mean score of 7.43. In terms of appearance. It was described as Moderately Firm having a mean score of 7.37, while in terms of taste, it was described as having a mean score of 7.3, and in terms of aroma, it was described as Moderately Inviting having a mean score of 7.23.

Table 5. Sensory Qualities of Treatment D-Cured Semi-dried Bangus in Batwan Flavor Considering Appearance, Texture, Taste, Aroma, and Its General Acceptability

Treatment D	Replicate			Total	Mean	Verbal Interpretation
	1	2	3			
Appearance	7.3	7.4	7.4	22.1	7.43	Moderately Tan
Texture	7.4	7.5	7.4	22.3	7.43	Like Very Much
Taste	7.1	7.4	7.4	14.5	7.37	Moderately Firm
Aroma	7.5	7	7.2	21.7	7.3	Moderately Inviting
General Acceptability	7.4	7.3	7.3	22	7.33	Like Moderately

The difference in the Sensory Qualities of the Four Treatments of Cured Semi-dried Bangus with Kamias, Libas, Batwan, and Labog Flavor

Data showed that there was no significant difference in the general appearance of the Cured Semi-dried Bangus with Kamias, Libas, Labog, and Batwan as flavors. The analysis of variance (ANOVA) showed an F-value of .330 having a 720 level of significance, which is very much higher than the .05 significance level. No significant difference was also found in terms of appearance, texture, taste, and aroma as shown by their F-value of .664, .760, .849, and .630 respectively, whose significance levels were also greater than .05.

Treatment A (with Kamias flavor) had the highest sensory quality among the four treatments in terms of appearance, texture, and taste, but Treatment A (with Kamias flavor) and Treatment C (with Batwan flavor) had both the highest in terms of aroma Treatment A had also the highest consumers' acceptability, although there were no significant differences that exist among the treatments considering all the variables such as appearance, texture, taste, and aroma.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 9 of 10

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

CONCLUSION AND RECOMMENDATIONS

Based on the findings of the study, the researcher is inclined to conclude that among the treatments, the best cured semi-dried boneless Bangus product for development is Treatment A (with Kamias flavor).

It is recommended that other variables should be considered in further experimentations on cured boneless Bangus and flavors. Technology on cured boneless Bangus and flavor should be transferred to students, food technologists, and homemakers.

(14) Output (Indicate the specific products, processes, or services that the project has produced; quantify when possible.)

Cured Semi-Dried (Hubag-hubag) Boneless Bangus: Potential for Commercialization

(15) Technology Transfer Activity (Indicate the specific products, activity design, processes, or services transferred as extension activities)

Production of Cured Semi-Dried (Hubag-hubag) Boneless Bangus: Potential for Commercialization

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI), and payback period.)

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(19) Literature Cited (include only those references published in reputable journals, and at least that from)

Devkota, H. P., & Sai, K. (2023). *Spondias pinnata* (L.f.) Kurz. Himalayan Fruits and Berries, 395-402.
<https://doi.org/10.1016/B978-0-323-85591-4.00042-8>

P. Vazhacharickal (2017). Phytochemical characterization of *Averrhoa bilimbi* and in vitro analysis of cholesterol lowering effect on fatty food materials. 31p. ebook.

Phytochemical characterization of *Averrhoa bilimbi* and in vitro analysis of cholesterol lowering effect on fatty food materials - GRIN

S. Quevedo et.al. 2014. Extraction and Characterization of “Batuan”. Department of Pure and Applied Chemistry.
https://annalsoftropicalresearch.com/wpcontent/uploads/pdf_files/Volume36No.2/5.pdf

Z. MiAmor. 2019. Development and Standardization of Libas Fruit (*Spondias pinnata*) as Base Seasoning Mix. University of the Philippines System. <https://www.studocu.com/ph/document/university-of-the-philippines-system/fundamentals-of-cropscience-ii/libas-study-lecture-notes-1/13599677>

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	
			Page	Page 10 of 10

H. Valentová, Z. Panovská. 2009. SENSORY EVALUATION | Taste. Encyclopedia of Food Sciences and Nutrition (Second Edition).

<https://www.sciencedirect.com/topics/nursing-and-health-professions/sour-taste>

<https://healthtips.ph/food/bangus-or-milkfish-benefits-dr-willie-ong-health-benefits/>